Alex Nikolaenko

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EDUCATION

MSc Financial Mathematics Sep 2026 — Jun 2027
London School of Economics and Political Science (LSE)

BSc Mathematics and Economics, First Class Honours 2023 — 2026
University of London, academic direction by LSE
Commendations 2023/24, 2024/25 · Algebra 96/100, top in the international cohort · Optimisation 91/100

Bachelor of Economics, Mathematics and Economics major 2022 — 2026
HSE University, International College of Economics and Finance (ICEF), Moscow
GPA 9.11/10, top 0.6% of cohort · 10/10 in Abstract Mathematics, Econometrics, Optimisation Theory, Time Series and Panel Data Analysis, Mathematics of Finance and Valuation

Moscow School No. 57 Mathematics and Economics class 2015 — 2022

RESEARCH

The GARCH-M Model with an Extreme-Downside Risk Premium 2026
with I. S. Slabolitskiy (HSE) · under review at *Economics Letters*

- Proposed an asymmetric GARCH-M model in which expected returns price ordinary and extreme negative shocks separately, with a closed-form unconditional variance.
- Estimated the extreme-shock threshold by chaining GEV and GPD fits on filtered residuals, cutting threshold RMSE by up to a quarter in Monte Carlo. On S&P 500 returns, downside risk is strongly priced.

BSc Thesis: Neural Networks vs Traditional Calibration of the SABR Model 2026
HSE Moscow · grade 9/10

- Trained a Sobolev-loss neural surrogate of the Hagan formula (Tanh, ELU, GELU, SIREN) and compared it with classical calibration on GLD options.
- The surrogate matched or slightly beat the classical fit with a better-conditioned Jacobian, and a neural warm start cut iterations in the Feb–Mar 2026 high-volatility window.

EXPERIENCE

Junior Business Analyst, B2B Lending Products Feb — Apr 2026
T-Bank, Moscow

- Designed the architecture of a B2B buy-now-pay-later lending system: gathered requirements from stakeholders, set and assigned development tasks, and kept the legal team updated on changes.
- Specified its accounting logic (commissions, accruals, payments), tariff system and MVP pricing, and its integrations with 1C and document-management platforms.

SELECTED PROJECTS

CS:GO / CS2 Skin Price Analysis Python · ARMA–GARCH · Cointegration
Volatility clustering, structural breaks and cointegration of co-moving pairs in in-game asset prices.

MOEX G-Curve Modelling Python · Short-rate Models · EVT/GPD · MLE
Vasicek, CIR and Hull–White calibrated to Russian government bonds by MLE, with GPD tail fits.

AWARDS & ACTIVITIES

- 2nd-degree diploma in Economics, "I Am a Professional" national student olympiad (2025).
- Two merit scholarships for the top 10 of the HSE cohort, incl. the ICEF Academia award (1 of 5 winners).
- Participant in A. N. Shiryaev's Financial Mathematics seminar, Moscow State University.
- Student mentor for first-years and voluntary Macroeconomics II tutor, HSE Moscow (2025–2026).

SKILLS

Programming Python (NumPy, Pandas, SciPy, PyTorch) · R · SQL · LaTeX · Git · Docker
Quant & ML Stochastic calculus · Volatility models · Extreme value theory · Deep learning · Optimisation
AI & Agents Claude Code · Agent-searchable Obsidian knowledge base with a custom retrieval skill
Languages Russian (native) · English (fluent)